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MILITARY SPECIFICATION

MOTOR GENERATOR PU-126( )/U
(ROTARY CONVERTER)

1. SCOPE

1.1 Scope.- This specification covers one type of rotary converter designated as Motor Generator PU-126( )/U. (See 6.4)

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents of issue in effect on date of invitation for bids form a part of this document to the extent specified herein.

SPECIFICATION

FEDERAL

FF-B-571	Bolts; Nuts; Studs; and Tap Rivets (and Material for Same)
QQ-S-781	Strapping, Flat; Steel
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-F-00320	Fiberboard; Sheet, Stock and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-B-131	Barrier Material, Water Vaporproof, Flexible
MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier Material
MIL-T-152	Treatment, Moisture and Fungus - Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric Acid Base)

FSC 6125

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MIL-P-11268	Parts, Materials, and Process Used in Electronic Communication Equipment
MIL-G-12803	Gasket Materials, Non-Metallic
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-E-55301	Electromagnetic Compatibility

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects For
MIL-STD-705	Generator Sets, Engine-Driven, Methods of Tests and Instructions

DRAWINGS

ELECTRONICS COMMAND

SC-DL-369300	Motor Generator PU-126( )/U Drawing and Data List

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS ASSOCIATION

S1.4-1961	General-Purpose Sound Level Meters, Specification For

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, New York 10016.)

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3. REQUIREMENTS

3.1 Description.- Motor Generator PU-126( )/U is a transportable rotary converter which converts three-phase, 60 cycle, 208 volt, alternating current to 400 cycle 120/208 volt, single or three phase, alternating current.

3.2 Preproduction samples.- The contractor shall furnish two preproduction samples for approval if required by the invitation for bids and contract. (See 4.3 and 6.2(d)(1))

3.3 Construction.- Motor Generator PU-126( )/U (rotary converter) shall be constructed in accordance with Drawing and Data List SC-DL-369300.

3.3.1 Parts, materials, and processes.- In addition to the requirements of this specification the requirements of Specification MIL-P-11268, including the selection requirements therein, shall apply.

3.3.2 Treatment of assembled equipment.- Parts and assembled equipment shall be treated in accordance with Specification MIL-T-152.

3.4 Castings.- Castings shall be of uniform quality and condition, and free from cracks, harmful shrinkage, porosity, gas holes, foreign matter, hard spots, and other injurious defects. The surface of the castings shall be free from pits, parting lines, porous areas, fins, ridges, nodules, raised metal and scale. Castings shall not be plugged or welded unless approval is granted by the procuring agency.

3.5 Cleaning.-

3.5.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.5.2 Equipment.- After assembly, equipment shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the equipment.

3.6 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

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3.7 Marking.- Marking shall conform to Specification MIL-M-13231. (See 4.4)

3.7.1 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.2 Serial numbers.- Each Motor Generator PU-126( )/U shall be serial numbered.

3.8 Riveting.- After riveting, the joints shall be tight; the joined parts shall be undamaged; and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.9 Soldering.-

3.9.1 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Welding.- Wherever practicable, welded joints shall be such that grinding on the finished weld will be unnecessary.

3.10.1 Cleaning prior to welding.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

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3.10.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum.

3.10.3 Cleaning after welding.- Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.11 Wiring and cabling.- Wiring and cabling shall be neat and sturdy.

3.11.1 Protection.- Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.11.2 Support.- Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire and cable (1) during and after subjection of the equipment to specified service conditions or (2) after service or repair of the equipment in a normal manner. Twine or tape shall not be used for securing wire and cable.

3.11.3 Clearance.- Clearance between wires or cables, and parts such as resistors shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.11.4 Connections.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

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3.12 Electrical requirements.-

3.12.1 Input.- Motor Generator, PU-126( )/U shall be capable of operating from a 60 cycle (±2 cycles), 208 volts (±20 volts), 3 phase alternating current source.

3.12.2 Power output (see 4.16).- The power output of the generator shall be as follows:

(a) Wye connection - 12.5 KW, 120/208 volts, 3 phase, 400 cycles ±3 percent, 4 wire, 0.80 power factor.

(b) Delta connection - 10 KW, 120 volts, single phase, 400 cycles ±3 percent, 2 wire, 0.80 power factor.

3.12.3 Voltage regulation (see 4.25).- The output voltage of the generator shall be maintained within the following limits for specified conditions:

(a) No load and any power factor from unity to 0.80 lagging - ±6 percent for 3 phase (wye connection) and ± 7 1/2 percent for single phase (delta connection).

(b) Any constant load to rated KW and any power factor from unit to 0.80 lagging - ± one percent.

(c) Following any sudden change in load between no load and rated KW and at any power factor from unity to 0.80 lagging, stable operating conditions shall be reestablished within four seconds. Stable operating conditions is defined as operation during which the voltage is in accordance with (b) above.

3.12.4 Overload protection (see 4.17).- The tripping time of the circuit breaker shall be as follows to protect the motor generator from overload:

(a) At 125 percent of rated load - 900 to 3600 seconds.

(b) At 200 percent of rated load - 70 to 120 seconds.

(c) At 300 percent of rated load - 9 to 16 seconds. In addition, the tripping time of the circuit breaker when the motor generator is subjected to a direct short circuit shall be such as (1) to prevent any damage and (2) to allow normal operation of the motor generator once the short circuit is removed.

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3.12.5 Wave form (see 4.27).- The wave form shall not contain any individual harmonic whose peak voltage is greater than 2.5 percent of the 400 cycle fundamental. The total harmonic content (square root of the sum of the squares), shall not exceed 2.8 percent of the 400 cycle fundamental. These requirements shall apply for both voltage connections and for any resistive load between no load and rated KW.

3.12.6 Voltage unbalance with unbalance load.- When "wye" connected for 120/208 volt, 3 phase, 400 cycle, 4 wire output, the voltage available from line to neutral shall not differ from any other line to neutral voltage by more than 5 volts, when tested as in paragraph 4.26.

3.12.7 Circulating current.- The circulating current shall gradually decrease to four (4) amperes or less as rated load is applied. (See 4.8)

3.12.8 Efficiency.- The overall efficiency of the motor generator shall not be less than 75 percent when tested as specified in paragraph 4.10.

3.12.9 Phase rotation.- The phase rotation at the output terminals, shall be L1, L2, L3, and N when the motor generator is turning in the direction of the rotation indicated by the arrow on the motor generator frame. (See 4.20)

3.12.9.1 Phase balance (voltage).- At the no load condition, the voltage of any phase shall not vary more than 0.2 volt from any other phase. (See 4.19)

3.12.10 Insulation resistance.- All windings shall have an insulation resistance not less than one megohm immediately after the temperature of the motor generator has been allowed to stabilize while carrying a rated load in an ambient temperature of 25°C, (see 4.12), except after subjection to the moisture resistance test.

3.12.10.1 Insulation resistance after moisture resistance test.- Following subjection to the moisture resistance test, specified in 4.15, the insulation resistance shall not be less than 50,000 ohms. (See 4.12)

3.12.11 Breakdown voltage.- Electrical insulation of the equipment shall withstand a minimum breakdown voltage of 1500 volts. (See 4.7)

3.13 Mechanical balance.- When tested as specified in 4.14, the maximum allowable total amplitude for balance shall be .00015 inch at 1800 rpm. Balancing shall be two plane dynamic.

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3.14 Overspeed.- Motor Generator PU-126( )/U shall be capable of withstanding an overspeed of 25 percent for 5 minutes without damage. (See 4.18)

3.15 Audio noise (see 4.6).- The noise level of the motor generator, when operating at any load from zero to rated full load, shall not exceed those specified as follows:

Frequency Band (Cycles per second)	Sound pressure level in decibels (0.0002 micro-bar reference)
20-74	75
75-149	75
150-299	70
300-599	65
600-1199	60
1200-2399	60
2400-4799	60
4800-10000	60

3.16 Suppression of radio interference.- The equipment shall meet the emanation and susceptibility limits of Specification MIL-E-55301 for Tactical equipment. Application of suppression components and techniques shall conform to that specification. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance with the specified emanation and susceptibility limits, he shall obtain approval from the contracting officer before proceeding with the modification. (See 4.21)

3.17 Service conditions.- The equipment shall meet the following service conditions — where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.17.1 Temperature (see 4.22).-

(a) Operating.- Ambient temperature in the range of +125°F to -65°F. (The 125°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating.- Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme no to exceed 24 hours, at any one time.

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3.17.2 Relative humidity (see 4.15).- Daily exposure up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.17.3 Elevation (see 4.11).-

(a) Operating.- Up to 10,000 feet above sea level.

(b) Nonoperating.- Up to 50,000 feet above sea level.

3.17.4 Dripproof (see 4.9).- The enclosure shall be of dripproof construction in accordance with section 3 of Standard MIL-STD-108.

3.17.5 Vibration.- With the equipment shock mounts blocked, the amplitude of vibration of any part, subassembly, or structural number of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.23)

3.18 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.13) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc.

3.19 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2(g) and 6.2.(h))

3.20 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein.

3.4 Castings
3.5 Cleaning
3.8 Riveting
3.9.1 Flux and cleaning agents for soldering
3.9.2 Process for soldering
3.10.2 Process for welding
3.10.3 Cleaning of welding
3.11.1 Protection for wires and cables
3.11.3 Clearance in wiring and cabling
3.11.4 Connections

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.28)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A and group B (see tables II and III, respectively).

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Table I - Preproduction inspection

Inspection (See 4.3 for additional preproduction inspection)	Req. Para.	Insp. Para.
Dripproof	3.17.4	4.9
Elevation	3.17.3	4.11
Moisture resistance	3.17.2	4.15
Phase rotation	3.12.9	4.20
Temperature	3.17.1	4.22
Vibration	3.17.5	4.23
Insulation resistance	3.12.10.1	4.12
Suppression of radio interference	3.16	4.21
Audio noise	3.15	4.6

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Finish	3.6
Marking	3.7

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government—at its discretion—may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6) Test equipment for Government verification shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to table II and the inspection procedures of Standard MIL-STD-105, using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government.

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Table II - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.20	4.24	2.5DPHU*	10 DPHU
Electrical				
Breakdown voltage	3.12.11	4.7	2.5 DPHU for the group	
Insulation resistance	3.12.10	4.12		
Power Output	3.12.2	4.16		
Efficiency	3.12.8	4.10		
Voltage regulation	3.12.3	4.25		
Wave form	3.12.5	4.27		

* Defects per hundred units.

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to table III and to the special procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection. The AQL for each inspection specified in table III shall be 6.5% defective; and the inspection level for normal, tightened, and reduced inspection shall be S-4.

4.5.2.1 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group B inspection

Inspection	Req. Para.	Insp. Para.
Circuit breaker performance	3.12.4	4.17
Circulating current	3.12.7	4.8
Interchangeability	3.18	4.13
Mechanical balance	3.13	4.14
Over speed	3.14	4.18
Voltage unbalance	3.12.6	4.26
Phase balance (voltage)	3.12.9.1	4.19
Phase rotation	3.12.9	4.20

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4.24.2.2 Preservation inspection. Inspection of preservation and interior

4.5.3 Group C inspection.- This inspection shall be as listed in table IV, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1.

Table IV - Group C inspection

Inspection	Req. Para.	Insp. Para.
Group I		
Elevation	3.17.3	4.11
Moisture resistance	3.17.2	4.15
Insulation resistance	3.12.10.1	4.12
Temperature	3.17.1	4.22
Vibration	3.17.5	4.23
Group II		
Audio noise	3.15	4.6
Group III		
Radio frequency interference	3.16	4.21
Dripproof	3.17.4	4.9

4.5.3.1 Sampling for group C inspection.-

4.5.3.1.1 Group I - Table IV.- One sample shall be selected at random, from each 200 motor generators produced or fraction thereof and subjected to each inspection specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.2 Group II - Table IV.- Two (2) samples shall be selected at random each month and subjected to the inspection specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.3 Group III - Table IV.- This inspection shall be required only when preproduction samples are not specified by contract or purchase order. When required, one sample shall be selected at random at the start of the contract from the first quality conformance inspection production lot (see 4.5.3), and subjected to each inspection specified. No subsequent inspection for the tests in this group will be required.

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4.5.3.2 Noncompliance.- The contractor shall immediately report in writing each group C failure, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective. (See 6.3)

4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Audio noise.- Measurements shall be made in a noise free area at a distance of three feet on both sides and ends of the motor generator and at a height of 5 feet above the base of the unit. At no point shall the sound level exceed that specified in paragraph 3.15. The measurements shall be taken using an octave band analyzer and a sound level meter conforming to the requirements of American Standards Association Standard S1.4-1961. Readings shall be taken with the flat overall frequency response setting of the instrument. This test shall be performed using an input voltage of 208 VAC, 60 cps, 3 phase.

4.7 Breakdown voltage.- The equipment shall be subjected to a breakdown voltage test in accordance with Method 302.1, high potential test, of Standard MIL-STD-705 to determine compliance with the requirement specified in paragraph 3.12.11.

4.8 Circulating current.- Motor Generator PU-126( )/U shall be tested for circulating current in accordance with Method 509.1 of Standard MIL-STD-705. (See 3.12.7) Input voltage shall be 208 VAC, 60 cycles, three (3) phase.

4.9 Dripproof.- The equipment, in its normal operating position, shall be subjected to the dripproof test specified in Standard MIL-STD-108. (See 3.17.4)

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4.10 Efficiency (see 3.12.8).- The overall efficiency of the motor generator shall be tested, using standard laboratory test equipment, at an input of 208 volts, 3 phase, 60 cycles, AC and a rated output load of 12.5 KW at 208 volts, 3 phase, 400 cycles AC, Wye connected. Efficiency to be calculated with the following formula:

$$\% \text{ Efficiency} = \frac{\text{Output (watts)}}{\text{Input (watts)}} \times 100$$

4.11 Elevation test (see 3.17.3).- The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
4.16	Generator Power Output 10.0 KW and 12.5 KW

(b) Reduce the barometric pressure to 20.6 ±0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.11(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ±0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.11(a). The equipment shall meet specified performance for these measurements.

4.12 Insulation resistance.- Insulation resistance test shall be performed on the equipment in accordance with Method 301.1 of Standard MIL-STD-705. (See 3.12.10 and 3.12.10.1)

4.13 Inspection for dimensional interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.18. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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(a) External and internal dimensions of cases, covers, insertable assemblies and replaceable parts when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.14 Mechanical balance.- Motor Generator PU-126( )/U shall be tested for mechanical balance using a Bear Balancing Machine, Model 400B, or equal. Meter magnitude reading of 10 on Bear Balancing Machine is equal to .00015 inch displacement. (See 3.13)

4.15 Moisture-resistance test (see 3.17.2).- The equipment shall be tested as follows:

4.15.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.15.2 Test method.-

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.15.3.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform

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to Standard MIL-STD-170. The measurements specified in 4.15.3 shall be made during the periods shown on the standard and shall comply with 4.15.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall meet full specified performance for those measurements specified in 4.15.3.

4.15.3 Performance.- Measurements made during test and during cycling shall be as follows:

Paragraph	Measurement
4.16	Generator power output

4.15.4 Failure.- If any equipment fails to meet the performance specified in 4.15.3 during cycling, it does not pass the test. In addition, if the equipment fails to meet specification requirements after conditioning as specified in 4.15.2(e), it does not pass the test.

4.16 Output.- Motor Generator, PU-126( )/U shall be connected to a 3 phase source and operated in both the wye and delta connections. The power output shall be tested, using standard laboratory equipment, to determine compliance with the requirements specified in paragraphs 3.12.1 and 3.12.2. This test shall be made using the following input voltages: 188 VAC - 58 cps, 208 VAC - 60 cps, and 228 VAC - 62 cps.

4.17 Overload.- The equipment shall be operated at 125, 200, and 300 percent of rated load to determine compliance with the requirement specified in paragraph 3.12.4. In addition, the Motor Generator Output shall be short circuted to check tripping dependability of the circuit breaker for prevention of any damage to the equipment. This test shall be performed using an input voltage of 208 VAC, 60 cps, 3 phase.

4.18 Overspeed.- Connect motor input to 208 volts, 3 phase power source, equipped with a variable frequency supply. The equipment shall be placed in operation at its rated input and the frequency adjusted until the no load speed is 125 percent of rated speed. Operate at this speed for 5 minutes and inspect in accordance with Method 505.1 of Standard MIL-STD-705. (See 3.14)

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4.19 Phase balance (voltage).- The equipment shall be tested for phase balance (voltage) in accordance with Method 508.1 of Standard MIL-STD-705 to determine compliance with the paragraph 3.12.9.1.

4.20 Phase rotation.- Motor Generator, PU-126( )/U shall be tested in accordance with Method 507.1 of Standard MIL-STD-705 for phase rotation to determine compliance with the requirement specified in paragraph 3.12.9.

4.21 Radio interference suppression.- The preproduction model equipped for the reduction of radio interference in accordance with paragraph 3.16 shall be tested, as described in MIL-E-55301, by the contractor or a commercial testing organization approved by the Government. The contractor shall furnish the contracting officer for approval the Electromagnetic Compatibility test report required in MIL-E-55301. The test report will include a block diagram of the test set up, a description of the interference reduction system applied, and a list of the test equipment utilized. Upon approval by the contracting officer, and provided all other requirements are met, the approved test sample shall be used as a visual standard for all production units.

4.22 Temperature test (see 3.17.1).- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Performance shall be measured as follows:

Steps	Paragraph	Measurement
1, 4, 7, 10	4.16	Generator power output

4.23 Test for internal vibration.- Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.17.5:

(a) Secure the equipment (shock mounts, removed or blocked) directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

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(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.24 Visual and mechanical.- Equipment shall be examined for the defects listed in MIL-STD-252.

4.25 Voltage regulation.- Voltage regulation of the motor generator shall be tested in accordance with Method 613.1 of Standard MIL-STD-705 to determine compliance with the requirement specified in paragraph 3.12.3. Input voltage shall be 208 VAC, 60 cps, 3 phase.

4.26 Voltage unbalance test.- Motor Generator PU-126( )/U shall be tested for voltage unbalance by applying a 10.8 ampere load from line to neutral. (See 3.12.6) Input voltage shall be 208 VAC, 60 cps, 3 phase.

4.27 Wave form.- The wave form of Motor Generator PU-126( )/U shall be tested in accordance with Method 601.4 of Standard MIL-STD-705. (See 3.12.5) Input voltage shall be 208 VAC, 60 cps, 3 phase.

4.28 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.29 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operation malfunction caused by the rough handling:

Insp. Para.	Measurement	Input voltage
4.16	Generator power output	208 VAC, 60 cps, 3 phase

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Each Motor Generator Set PU-126( )/U shall be packaged in accordance with item 1 through 7 of Figure 1 and Table V.

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MOTOR GENERATOR PU-126/U

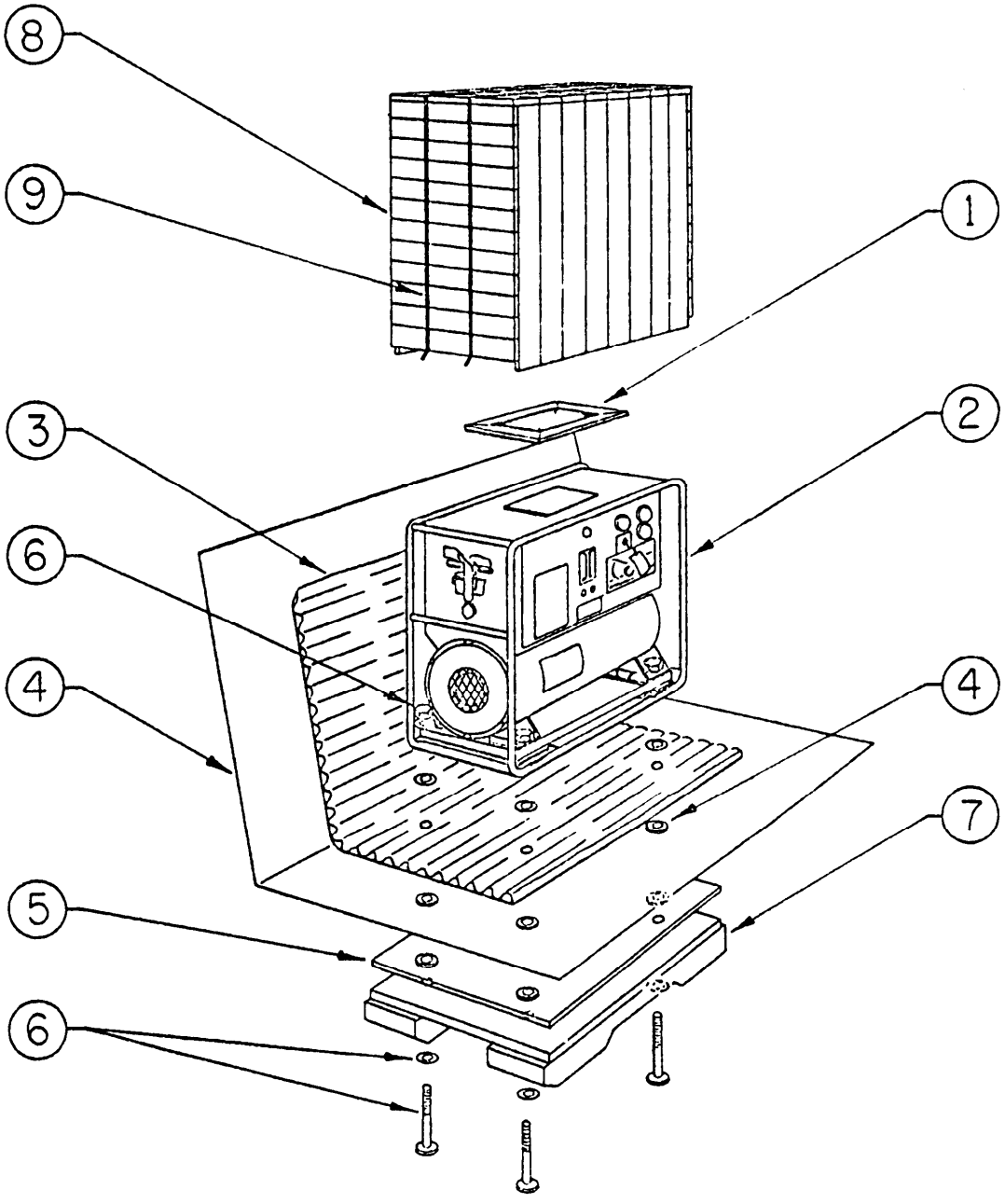


FIGURE 1

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BILL OF MATERIAL
Table V

RFP No.	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENTS Spec No.	T	G	C	S	Flute	PSI
1	Tech. Manual	2	1C-1										
	Bag, Barrier	2		9-1/2	7-1/2		MIL-B-117	11		d			
2	Motor Generator Set PU-126()/U	1	1A-16	32-5/8	19	34-1/4	Bare weight - 696 lbs						
	Blocking, Wood	4		3-1/2	2-1/2	1-1/8	PPP-B-621						
			Place two blocks at each end				under unit and between the shock mounts						
3	Cushion	1		112	64		PPP-P-291	11		2			
4	Barrier Heat Seal	1		118	60		MIL-B-131			1			
	Gaskets	8		3	3	1/16	MIL-G-12803	111	4	2	Group 3		
	Adhesive						MIL-A-140	11	8				
5	Pad, Bottom	1		36	22-1/2	1/8	PPP-F-00320	1	5	3	Variety A		
6	Bolts with Nuts and Plain Washers	4		4	3/8 Dia		FF-B-571						
7	Skid Base Made From	1					PPP-B-621						
	Skid, Floor	2		40-3/4	8	2							
				1 inch boards Made to fit Item 8									
8	Box, Wood	1		37-5/8	24	36-1/2	PPP-B-621			2	7		
9	Strapping, Metal	2		172	3/4	.023	QQ-S-781	1	2	B			

Packed Weight - 897 lbs
Packed Volume - 27.63 Cu. ft.

LEGEND
T - Type S - Style
G - Grade C - Class

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5.1.2 Level C.- Each Motor Generator Set PU-126( )/U shall be packaged as specified in 5.1.1 except item 4 shall be omitted and item 5 shall be class 2.

5.1.3 Package performance.- Package testing shall be performed in accordance with the requirements of Specification MIL-P-116. The rough handling test and cyclic exposure test will be required only when invoked in the bid request or contract. (See 6.2(2))

5.2 Packing.-

5.2.1 Level A.- Each Motor Generator Set PU-126( )/U shall be packed in accordance with items 8 and 9 of Figure 1 and Table V . The box closure shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.- Each Motor Generator Set PU-126( )/U shall be packed as specified in 5.2.1, except item 8 shall be class 1.

5.2.3 Level C.- Each Motor Generator Set PU-126( )/U shall be packed in shipping containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptancy by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the mode of transportation.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Motor Generator, PU-126( )/U is intended primarily as a source of power for the operation of Radar Surveillance Center AN/GSS-1, Radio Set AN/TPS-1D, and similar electronic equipment. It may be used anywhere that a suitable souce of 208 volt, 60 cycle, three phase power is available.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

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